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Review

On waveform distortion in the frequency range of 2 kHz–150 kHz—Review and research challenges

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ABSTRACT

The frequency range between 2 and 150 kHz has recently gained significant attention, triggered by standardization needs and increased emission in this wide frequency range. This paper gives an overview of the state-of-the-art concerning these so-called supraharmonics, and noticeably indicates the research challenges associated with waveform distortion in this frequency range, with emphasis on the following aspects: emission; propagation; interference; measurements; standardization; modelling and simulation.

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1. Introduction

Distortion of voltage and current in the frequency range between 2 and 150 kHz is not a new topic; engineers and manufacturers of end-user equipment have known about these emissions and solved problems associated to it, case-by-case for decades [1], but the amount of systematic knowledge and the dissemination of this have been limited. Recently, this frequency range has received more attention by researchers [2,3], and especially by standard-setting organizations [4,5]. There are considerable ongoing activities within IEC (both SC 77A and SC 205A), CENELEC and IEEE to develop standards, e.g. compatibility levels, limits and adequate test methods, covering this frequency range. Some standards do exist already [4] but there is no accepted framework, as there is for lower frequencies. The recent standard development was mainly triggered by the use of this frequency range for power-line communication (PLC) as part of smart metering infrastructure. An additional driving force for research is the increasing amount of equipment with active switching resulting in non-negligible emission in this range. Among others as input to the standardization work, a better understanding is needed of conducted phenomena in this frequency range.

The aim of this paper is to summarize the state-of-the-art and to give guidance to those new to the subject of waveform distortion between 2 and 150 kHz and to indicate research challenges and opportunities for future research.

Section 2 of this paper gives a brief overview of the terminology on waveform distortion. Sections 3 and 4 give the state-of-the-art on emission and propagation in the frequency range 2–150 kHz. Section 5 presents known cases of interference. Sections 6–8 summarize the measurement, standardization and modelling issues. The important research challenges, extracted from the other sections, are summarized in Section 9, followed by the conclusions in Section 10.

2. Terminology on waveform distortion

The frequency range up to 150 kHz contains different classes of distortion. Based on the frequencies involved, the following classes are distinguished:

- Components at integer multiples of the power-system frequency. Waveform is distorted but periodic with the same period as the power-system frequency. These components are referred to as “integer harmonics” or “harmonics”.
- Components at non-harmonic frequencies, (interharmonics).
- Components at frequencies less than the power-system frequency (subharmonics). These are also a subclass of interharmonics.
- The component at zero frequency (dc).
- Components at any frequency between 2 and 150 kHz, referred to as “high-frequency harmonics” [6,7], “high-frequency distortion” [8] or more recently “supraharmonics” [4].

The term “low-frequency emission” is used within IEEE EMC society to refer to frequencies below 150 kHz. Within IEC the term “low frequency” refers to frequencies below 9 kHz. The terminology used for any waveform distortion does not consider the origin or impact of the distortion. There is for example no sudden change in physical character of the distortion at 2 kHz that would justify that as the upper border for harmonics. Also, for example, different causes of interharmonics exist, but regardless of the origin, any component at a non-harmonic frequency will be referred to as an interharmonic. The above terminology also does not consider the time-varying nature of the distortion [9,10,11].

Although the term “supraharmonics” is being used more frequently, the term is still new to the electric power community. But the authors still consider this as the most suitable word available at the moment. The authors therefore recommend using the term “supraharmonics” to refer to all quasi-stationary waveform distortion in the frequency range 2–150 kHz.

The term “supraharmonics” as proposed here, only covers quasi-stationary phenomena in the same way as the terms “harmonics” and “interharmonics”. The frequency range between 2 and 150 kHz also contains transient phenomena, but a further discussion of these is beyond the scope of this paper.

3. Emission

3.1. Types of emission

When considering the frequency domain, principally two types of emission can be considered: narrowband and broadband. In general, a spectrum that contains a distinct single frequency component is considered narrowband while a spectrum with several adjacent components is considered broadband. The bandwidth of the components is depending on the time resolution of the signal, window length, window type, and grouping method (i.e. a 2 kHz grouping might show a narrowband signal, where a 200 Hz grouping shows a broadband). To obtain a spectrum, a certain time window is needed; time variations within this window can blur the distinction between narrow and broadband. A time-varying signal can be narrowband over a short time window but broadband when considering a longer period. It was shown in Ref. [12] that what appeared as broadband emission between 40 and 80 kHz when performing a DFT over a 200 ms window was in fact narrowband emission at changing frequencies with a duration of just a few milliseconds. This phenomenon is due to the variation in

switching frequency with hysteresis control, as described in Ref. [13]. The emission from the two main sources of emission above 2 kHz, residues from power–electronic converters and transmitters of power-line communication can be classified as either narrow or broadband emission.

The work presented in Refs. [2,12] has shown that, when analysing in the time domain, an additional type of emission can be distinguished: short-duration oscillations, with a frequency of a few kHz, that appear twice every cycle of the power-system frequency. The oscillations are part of the so-called "zero-crossing distortion". This phenomenon is known from the design of power–electronics converters [14,15] and consists of two different parts:

- A brief period around the voltage zero-crossing during which the current is zero, referred to as "leading-phase distortion" in Ref. [16].
- An oscillation in current, with a frequency of a few kHz, at the end of this period, "ringing distortion" [16] or "zero-crossing oscillation" [2].

It is the latter one that contributes to the emission in the frequency range 2–150 kHz. For instance in Refs. [16,17] the zero-crossing oscillations are shown to be amplified by the EMC filter. Looking at this type of emission in the frequency domain, recurrent oscillations could also be characterized as either narrow or broadband. They do however differ in the sense that, when seen in the time domain, they appear as a damped oscillation and always around the zero-crossing of the voltage. Another reason for emission in the frequency range above 2 kHz is simply the continuation of the emission below 2 kHz; typically odd harmonics of decaying magnitude with increasing frequency. Especially sharp changes in voltage or current magnitude, as with commutation notches [18], give rise to such emission. As with recurring oscillations, this emission would also appear as either narrow or broadband emission in the frequency domain.

3.2. Sources of emission

The two main sources of emission above 2 kHz, that have been identified, are power–electronic converters with active or passive switching and transmitters of power-line communication. With some exceptions, the latter are of much higher magnitude than the former ones. However, power-line communication is typically present during short periods of time, whereas emission from active converters is often present for a substantial part of the time. The arrangement causing the emission from among others PWM inverters is described in detail in Ref. [19]. The spectrum of a PWM switching pattern is concentrated around the switching frequency and integer multiples of the switching frequency. As mentioned in Section 3.1, also some passive converters emit in this frequency range, but their main emission is at lower frequencies.

One of the reasons for the use of active converters is the ability to significantly reduce the emission at lower frequencies and in this way to comply with emission limits. There are other advantages of active converters however, so that loosening of emission limits at lower frequencies will not in all cases result in a reduction of the emission at higher frequencies. It is however important to consider this relation in the discussions about emission limits.

Emission in the frequency range between 2 and 150 kHz is shown for different devices by several authors. Emission around 3 kHz and multiples of this, due to the switching frequency of a 1 MVA PWM converter is shown in Ref. [18]. In Ref. [20] it is shown that emission from a PWM boost converter can be found in the frequency range 2–9 kHz. Emission in the range 9–150 kHz from different industrial-size converters is presented in Ref. [20]. Several examples of emission in the range between 2 and 150 kHz are also shown in Ref. [5].

Emission around 6 kHz, due to oscillations around commutation notches, at a LV bus is shown in Ref. [18]; the range of such oscillations is up to 10 kHz.

Frequencies up to several tens of kHz are measured for LED street lamps and for high-pressure sodium lamps with electronic ballast [25]. For electromagnetic ballast, only decaying harmonics up to 20 kHz are visible. In the frequency range 2–9 kHz, the emission with electromagnetic ballast is about 2.5 times the emission with electronic ballast. For frequencies above 9 kHz, the electronic ballast shows higher emission. Current emission associated with, among others, low power LED lamps are discussed in Ref. [26], where it is shown that the emission seems to be organized in bands.

Emission from a number of electric vehicle chargers is presented in Ref. [27]: the emission was mainly concentrated in the range up to about 15 kHz, but one of the chargers showed an emission peak somewhat above 30 kHz. Emission at frequencies 27 kHz, 37 kHz and 100 kHz is presented in Ref. [28] and at 21 kHz in Ref. [29]. Analyses of the emission from power converters connected directly to the power grid in an experimental microgrid composed by PLC equipment and several PV inverters with a switching frequency of 16 kHz, appearing in the voltage when the PV inverter is operating, are presented in Ref. [30].

Emission at 4 kHz and 16 kHz from PV inverters is shown in Ref. [31]. Emission in public low-voltage grids, due to PV inverters, is shown in Ref. [32]. The main emission is shown to be in the range 16–19 kHz, with 16 kHz the most common one. Voltage distortion up to about 1 V was measured; levels were clearly higher at daytime, but there was no correlation with the amount of installed capacity. Voltage distortion was lower in grids without PV and there was a wider range in dominant frequencies; 6.2–28.8 kHz. It is shown, experimentally, that there is emission from PV inverters in the range between 16 kHz and 17 kHz in [33]. Reference [34] shows an increase in voltage distortion at 16 kHz, 20 kHz and 32 kHz due to PV. Similar results are presented in Ref. [35] where measurements from single phase PV inverters are shown to cause distortion around 16 kHz. In Ref. [35] measurements are also presented from larger solar PV plants, showing the typical spectra associated with an inverter using PWM [36] with components around 4 kHz. Emission from an 850-kW solar park is shown to have peaks around 3 kHz, 6 kHz and 9 kHz [37]. Measurements in a 1-MW solar park with a mixture of small and large inverters are shown in Ref. [38]; emission occurs at 3 kHz, 6 kHz and 9 kHz due to large inverters and at 10 kHz, 16 kHz and 25 kHz due to small inverters. In Ref. [39], PV inverters employing PWM injecting current in the range of 2–5 kHz are discussed. A case study presented in Ref. [40] shows the switching frequency from a grid interface inverter of a variable speed wind turbine located at 3.15 kHz.

According to Ref. [41] the peak voltage can be up to 50 V in a 100-V low-voltage network due to resonance together with emission. High levels of voltage distortion, "up to several volts", in the kHz range due to a washing machine and a cell tower are shown in Ref. [42].

Emission from a number of low-voltage devices (i.e. compact fluorescent lamp; CRT television; vacuum cleaner with brush motor) is presented in Ref. [43] as noise experienced by power-line communication. Similar measurements, but performed in the early 80 s are shown in Ref. [1]; the main sources according to that study where universal motors and the line frequency of television receivers.

It is shown in Ref. [21] that slot harmonics from squirrel-cage induction machines can appear between 1 and 1.7 kHz. In Ref. [22] these are referred to as "tooth ripples". The slots in synchronous and induction machines were already mentioned as a source of harmonics in Ref. [23] and later in Ref. [24]. Note that the presence

of slots also causes emission at integer multiples of the frequency with which the air-gap field passes the slots, which will be in the range above 2 kHz. It is shown by simulations in Ref. [21] that the magnitude at the second and third harmonic of the "slot frequency" is close to the magnitude at the slot frequency.

3.3. Reducing emission

Emission can be reduced in a number of ways, where the principal distinction is between the following two approaches:

- Changing the switching pattern and control algorithms so that less emission is produced during the power–electronics switching.
- Adding a passive filter between the actual converter and the grid [44,45,46] .

One way of utilizing the first approach would be the use of multi-level converters that offer a range of possibilities for reducing emission around the switching frequency and its harmonics [24,31]. The above-mentioned passive filter is known as "EMC filter" or "EMI filter" and is a common part of most small devices. Its aim is mostly to reduce emission in the frequency range above 150 kHz, but it can also be used to reduce emission in the range 2–150 kHz. The latter would however require a redesign of the filter, avoiding resonances below 150 kHz.

4. Propagation

Using the superposition principle for linear systems, the voltage (at frequency f) at a bus r due to sources at different busses, is found from the following expression:

$$U_r(f) = \sum_{s=1}^{N} Z_{rs}(f) \cdot I_s(f)$$

where $Z_{rs}(f)$ is the transfer impedance from bus s to bus r. Nonlinear phenomena are not included in this expression, but there is simply insufficient knowledge available on how these will impact the propagation of distortion in the range between 2 and 150 kHz.

All parameters in the above expression are complex numbers and are varying with time. The presence of an additional device and variable load condition further impacts the transfer impedance between neighbouring devices. It is therefore not possible to simply add the emission from individual devices and aggregation rules as in IEC 61000-3-6 do not hold anymore, as was shown for example in Refs. [47,48,52]. Next to classical phenomena, like diversity between the emission spectra from individual devices, diversity in time of use of equipment, and cancellation of emission in the complex plane, also the time variation of the transfer impedance should be considered.

Propagation is studied in this section from three different angles: propagation from one device to one or more neighbouring devices and/or to the rest of the grid; resonances either in the installation or in the rest of the grid that result in an amplification of the transfer; and the receiving by a device of the emission propagated from elsewhere. The source of the emission, whether it is a device like a converter or a transmitter of power line communication, is of no relevance with regard to propagation. What matters are the transfer impedances between the nodes in the grid at the instance when the emission is injected into the system.

4.1. Propagation between devices

Propagation between devices in a low-voltage installation has been studied in Ref. [47]. It was shown that the current at the supply terminals of the installation was much smaller than at the device terminals. By measuring with all connected devices it was found that the current flowed mainly between devices and not towards the grid (see Section 4.3).

The currents in a low-voltage installation with 1–48 fluorescent lamps with electronic ballast were measured in Ref. [48]. It was shown that the emission at tens of kHz, from the complete installation, reduced with increasing number of lamps. This behaviour has been explained in Ref. [52] using a rather simple model: a constant-current source in parallel with a capacitor representing the EMC filter. It is this capacitor that creates a low-impedance path causing the current to flow mainly between devices. This low-impedance path is also responsible for some of the interference between devices and power-line communication [3,53]. Measurements in a low-voltage distribution network [32] showed between 1 and 2 V distortion at 8 kHz at 300 and 400 m from the distribution transformer; however, the voltage at that frequency was close to zero near the distribution transformer. The propagation of emission originating from distributed generation is also discussed in Refs. [30,39].

Ref. [48] shows a different behaviour for the zero-crossing oscillations at a few kHz: their amplitude increases with increasing number of devices and the resonant frequency gets somewhat smaller. No quantitative explanation has been given yet for this phenomenon.

4.2. Resonances

Resonances play an important part in the propagation of emission in the frequency range between 2 and 150 kHz. Emission close to the resonant frequency will be amplified when insufficient damping is present. Simulations for a number of wind parks [54] show that the emission from individual turbines around the switching frequency can be amplified at the point of connection. Those studies, as well as Ref. [55] show that the emission of the park as a whole becomes negligible above the main resonant frequency.

An extreme example of amplification is shown in Ref. [56] where the voltage at a 30 kV cable terminal contains an oscillation around 2 kHz and a magnitude between 30 and 100% of the fundamental voltage. The resonance in that case was due to underground cables in an industrial installation. In Ref. [57] measurements and simulations are shown as a resonance around 6.7 kHz between a 500-m cable and the power transformer.

In low-voltage installations, where the propagation is mainly between devices, the capacitance and inductance of those devices is also expected to play an important role. Connecting or disconnecting a single device can change the resonant frequency and therewith the propagation of the emission.

Simulations presented in Ref. [58] show that the length of the cable between an emitting device and the distribution transformer can have a strong impact on the 35-kHz voltage on the LV side of the transformer. For a cable length of 800 m the voltage is amplified by a factor five. It can thus not be generally concluded that disturbance levels get smaller when moving away from the source, as was suggested in Ref. [32]. It was further shown in Ref. [58] that there are no simple relations between grid parameters and resonant frequencies. Resonances in the frequency range 2–9 kHz are studied in Ref. [41], especially with reference to the size of the capacitor representing neighbouring equipment.

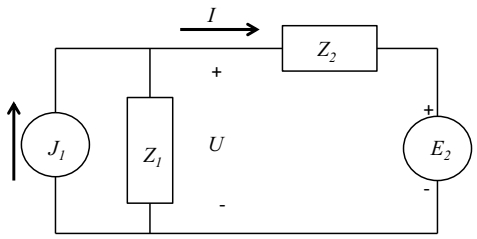


Fig. 1. Primary and secondary emission.

4.3. Secondary emission

The terms "primary emission" and "secondary emission" were introduced in Ref. [52] and are further explained in Ref. [59]. With reference to Fig. 1, the primary emission (both voltage and current) is driven by J_1 and the secondary emission by E_2; the current I at the device terminals is the sum of primary emission I_1 and secondary emission I_2:

$$I_1 = \frac{Z_1}{Z_1 + Z_2} \cdot J_1 \tag{1}$$

$$I_2 = -\frac{E_2}{Z_1 + Z_2} \tag{2}$$

$$I = I_1 + I_2 = \frac{Z_1 \cdot J_1 - E_2}{Z_1 + Z_2} \tag{3}$$

Similar expressions can be obtained for the voltage. From the above expressions it follows that both primary and secondary emission are dependent on the source (Z_1) and device impedances (Z_2). The model in Fig. 1 is only aimed at illustrating the concept, not as a tool for detailed studies. Especially close to device terminals, non-linear phenomena like cross-over modulation between different frequencies will play an important role. A third phenomenon, referred to as "interaction" has recently been proposed by an international working group to cover some of the non-linear phenomena [59].

An important conclusion already from this simplified model is that emission measurement may not be reproducible at another location as both E_2 and Z_2 are likely to be different. Such observation has been confirmed by measurements presented in several publications [28,31,58,60,61]. It should be noted that the presence of a distorted voltage at the terminals of the device (U in Fig. 1) might impact the internal emission J_1; this will also result in differences in emission at different locations. Distinguishing between these impacts remains a major challenge. In one of the cases presented in Ref. [28] not only the magnitude of the emission changes, also the main emission frequency changes (from 37 to 35 kHz). Measurements presented in Ref. [61] show that the primary emission from a PV inverter is strongly dependent on the presence of other devices in the same low-voltage installation.

A further consequence of secondary emission is that it is often difficult to interpret measurements of emission like those presented in Section 3.2.

The terms "primary emission" and "secondary emission" are new, but the underlying concepts have been around for a long time. A range of methods has been proposed to distinguish between different harmonic contributions, e.g. at the point of connection of an industrial installation [49,50,51].

With reference to Fig. 1, only U and I can be obtained from measurements. Without any additional assumptions it will not be possible to obtain values of the two sources (J_1 and E_2) and of the two impedances (Z_1 and Z_2). Note that for the general case, none of the values can be assumed constant in time. The different approaches proposed in the literature to distinguish between primary and secondary emission, differ through the assumptions made. All earlier work on this subject has been directed towards lower-order harmonics, typically the 5th and 7th order. It is not clear to which extent the assumptions made are valid for the frequency range above 2 kHz.

5. Interference

Several examples are shown in the literature of interference due to voltage and current distortion in the range of 2–150 kHz. The reported cases of interference can be grouped into four categories with Category 1 the most-commonly reported one:

- A device is occasionally not operating as intended or with reduced functionality.
- Failed operation or damage of a device.
- Interference with or due to power line communication.
- Audible noise from a device or installation.

The following reported cases can be grouped into Category 1 (a device is occasionally not functioning as intended or with reduced functionality):

- Reduced quality of the output of a copy machine due to frequencies in the lower kHz range [18].
- Control problems with equipment using a 77.5 kHz radio signal for internal clocks. Radiated emission at a multiple of the 3 kHz switching frequency of a drive interfered with the clock signal [18].
- The possible interference between a VSC HVDC converter and the 77.5 kHz radio signal [63].
- Cases of interference with light dimmers, medical equipment, electricity meters, and earth leakage current breakers [5].
- An 8-kHz voltage of about 5 V, due to a CNC mill, resulted in a range of complaints [32]: malfunction of a fully automated coffee maker; a hair dryer randomly turning on an off; malfunctions of the control of the CNC drive itself.
- A case is described in Ref. [62] where the emission from a large rectifier causes mal-functioning of the dimming with LED lamps in nearby offices. The mal-functioning was reproduced in a laboratory environment and was shown to be due to oscillations in the kHz range.
- Unwanted tripping of earth leakage current breakers is the main impact of emission in the range 2–9 kHz in Japan [41].
- Voltage distortion in the range 2–150 kHz was shown to effect light output from LED lamps in Ref. [64]

The following reported cases can be grouped into Category 2 (failed operation or damage of a device):

- Regular failure of small drives due to about 1% oscillations in voltage in the kHz range [18].
- Violent failure ("black smoke") of a precision measurement instrument [18].
- Failure of varistors due to repetitive activation by recurrent oscillations in voltage [18].

The following reported cases can be grouped into Category 3 (interference with or due to power line communication):

- Several cases of interference were presented in Ref. [62] including mal-functioning of touch-dimmer lamps and problems with data transmission from smart meters.
- According to Ref. [53], the major causes of PLC issues are the noise level along the lines (generated by loads connected to the grid) and low impedance of specific loads or line attenuation.

- Several examples of interference are shown in Ref. [5], where the emphasis is on interference to power-line communication.

The following reported cases can be grouped into Category 4 (audible noise from a device or installation):

- Telephone interference used to be an important consequence of high harmonic currents, especially in the frequency range 2–4 kHz, where the human hearing is most sensitive [22].
- Voltage distortion around 12 kHz was shown to cause mechanical oscillations and audible noise [18].
- Audible noise coming from a television is mentioned in Ref. [32].
- Audible noise from an inverter-driven motor, in the kHz range is reported among others in Refs. [65,66]. The role of mechanical resonances is shown to be very important with the generation of audible noise. With these examples, the motors are exposed to waveforms with high levels of distortion.

Many more examples appear in the informal literature and in the form of personal communication. Especially for the latter examples, it is difficult to verify if the interference is really due to disturbances in the frequency range.

6. Measurements

6.1. Measurement accuracy

Obtaining sufficient measurement accuracy for voltages and currents in the frequency range of 2–150 kHz is not trivial and has to be considered for the whole measurement chain. It requires transducers that have high accuracy for both amplitude and phase angle over a wide frequency range. In addition to appropriate anti-aliasing filters the measurement signals (voltage or current) need to be high-pass filtered in order to avoid leakage effects. The high-pass filter can be digital for A/D resolution of 16 Bit and better [68], otherwise an analogue filter is needed.

A detailed description of measurement techniques for waveform distortion in the frequency range of 2–150 kHz is given in Ref. [69]. Challenges for measuring and analysing such distortion are also discussed in Refs. [71,72]. In Refs. [68] two measurement methods are discussed regarding the necessity of filters, chosen bandwidth, impact of band boundaries and aggregation in time.

6.2. Signal processing analysis

Measurements show that signals in the frequency range from 2 to 150 kHz often change in amplitude and frequency on timescales of milliseconds [2]. A similar phenomenon is shown in Ref. [73] but for a much higher frequency band. Therefore both time-domain and frequency-domain should be considered for signal analysis. A joint time–frequency domain analysis like the short-time Fourier-transform (STFT) or sliding-window ESPRIT [74] is an appropriate tool to analyse these changes. With the STFT a signal is divided into smaller windows (with or without overlap) for every one of which a spectrum is obtained. These time-dependent spectra are combined to create an image of the changes in the spectrum over time (the spectrogram) [69]. The STFT is equivalent to applying a set of band-pass filters [75]. A broadband signal is decomposed into a set of time-varying narrowband components at different frequencies. This gives a lot of additional information compared to the spectrum obtained over a longer time window.

For the recurrent oscillations, neither the conventional spectrum, nor the spectrogram is an appropriate tool. A time-domain analysis is most convenient to quantify and study recurrent oscillations, like in Ref. [69], where ESPRIT is used to estimate frequency and amplitude of recurrent oscillations. In Ref. [70] a method for time–frequency analysis is proposed that is shown to give detailed information in term of frequency, magnitude as well as time-variability.

6.3. Standardized measurements

IEC 61000-4-19 defines immunity requirements and test methods for equipment to disturbances in the range of 2–150 kHz. A measurement method for the range of 2–9 kHz is given in IEC 61000-4-7. Measurements are also defined in the CISPR standards where typically heterodyne frequency-domain methods are used. IEC 61000-4-30 includes informative annex C describing measurement methods for 2–9 kHz as well as 9–150 kHz. CENELEC study report II devotes one section on measurements [5].

7. Standardization

A key term in the IEC standardization on electromagnetic compatibility, where the frequency range of 2–150 kHz is part of, is the “compatibility level”: “The specified electromagnetic disturbance level used as a reference level for co-ordination in the setting of emission and immunity limits” [76]. Compatibility levels are important in order to define compatibility margins for emission and immunity as a basis to achieve electromagnetic compatibility with a high probability.

However, after intense discussions in the relevant study committees in recent years, consensus has been reached only in the frequency range of 2–30 kHz [77]. Until now, a compromise for non-intentional emissions has not been reached for the range 30–150 kHz [78,79]. There are two different proposals of compatibility levels for non-intentional emissions between 30 and 150 kHz; one (“proposed compatibility level” in Fig. 2) being supported by experts related to equipment producing industry excluding industry providing PLC and another (“Compatibility level option B”) being supported by experts related to electricity distribution, plus PLC industry experts.

Fig. 2 further shows the voltage characteristics for “mains signalling” (2–95 kHz) according to EN 50160 as well as the emission limits for communication in the communication band and in non-communication bands according to EN 50065. The limits for the communication band are the same as in IEC 61000-3-8.

Following a recent SC 77A plenary meeting [80] it was decided to modify maintenance work for standard IEC 61000-2-2 into a new amendment instead of a standard revision. It was also decided to focus on frequency range of 2–30 kHz, while keeping 30–150 kHz under consideration. As consensus has been reached for compatibility levels in this frequency range, the present IEC target now is completion of standard maintenance work. Finally, maintenance work item for IEC 61000-2-12 (medium voltage) has been cancelled due to lack of agreement on compatibility levels on the whole frequency range in IEC 61000-2-2 (low voltage). Input to the standardization committees from an independent research community is needed to finalize the work.

8. Modelling and simulation

The modelling efforts in the frequency range of 2–150 kHz have to be considered for two distinctly different aspects; the grid and the components/devices/installation connected to the grid. Grid modelling for this frequency range (Section 8.1) is a well-developed area, especially for overhead lines and for medium-voltage and high-voltage cables, although knowledge on low-voltage installations remains limited. Device modelling (Section 8.2) still requires a lot of further studies.

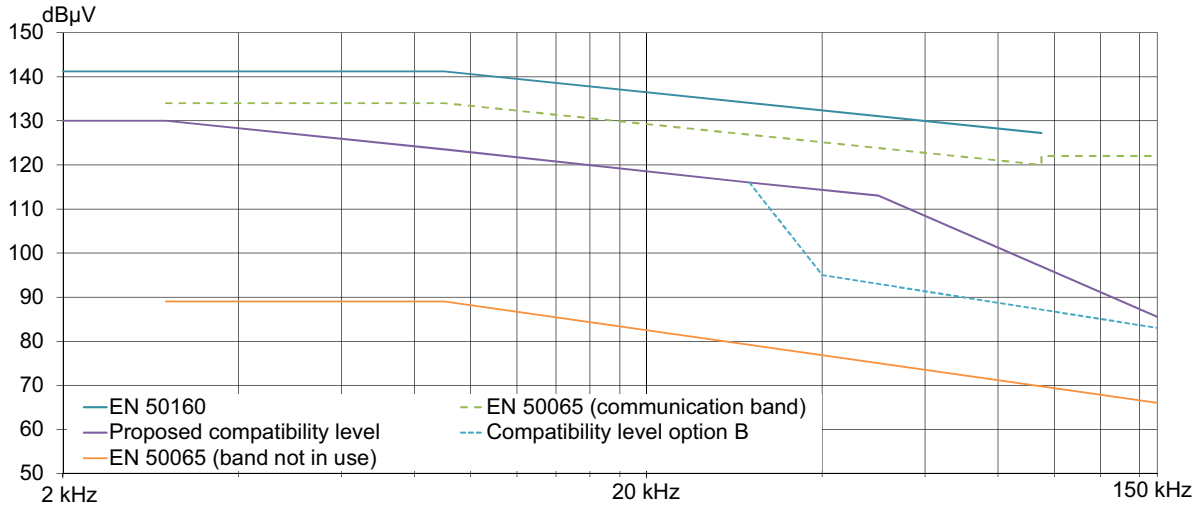


Fig. 2. Compatibility levels proposed by two different IEC expert groups [78] and other relevant limits [4].

8.1. Grid modelling

The modelling of the grid for this frequency range has strong similarities to the modelling for transient overvoltage as part of insulation coordination [81]. Looking into studies on transient overvoltages there are several papers on high frequency behaviour of transformers [82–85]. The drawback with all these models is that measurements are needed on each transformer to obtain the model parameters. General guidelines on transformer modelling are given in Refs. [86,87], where the different capacitive couplings are shown to be important. The models of the inductive part of the transformer, including the core losses, may lose validity at a frequency of several kHz. It is however also at these frequencies where the capacitive coupling will become dominant, so that the model inaccuracy becomes less relevant.

Models of cables and overhead lines can be found in the literature on power line communication e.g. Ref. [88] as well as in the literature on switching transients and EMTP modelling [86,89]. The basic expressions for overhead-line modelling were derived already around 1925, almost simultaneously by Refs. [90,91], but the approximations presented in Ref. [90] enabled them to be implemented in computer calculations, like the line-constants routine that formed the input for the overhead-line model used in the EMTP program [92]. It was shown in Ref. [93] that the expressions used were mathematically incomplete but of sufficient accuracy up to several hundreds of kHz. Simulation results were verified by measurements in Refs. [94,95]. A modelling study of MV lines with specific reference to harmonics in the frequency range 2–150 kHz is presented in Ref. [96].

The modelling effort for overhead lines, cables and transformers has been directed mainly towards the transmission system instead of the distribution system. Models for overhead lines and cables remain valid for lower voltage levels, but it will not be possible to perform measurements on each distribution transformer. This is where a modelling effort is still needed. In Ref. [97] the model of a transformer for power-line communication is presented. It shows the first resonances to be around 300 kHz. In Ref. [98] a modelling approach based on the transfer matrix of the transformer is presented. Measurements of the transfer for a 100 kVA, MV/LV transformer are presented in Ref. [99]: the transfer from LV to MV is amplified by a factor of 10 at the resonance frequency around 4 kHz; the transfer from MV to LV remains about the same as at the power-system frequency up to at least 80 kHz.

Especially for an LV installation one has to consider all the components within the installation, connected equipment as well as wires. Wiring inside a building is neither symmetrical and currents not balanced nor as structured as overhead lines and cables.

In summary, for the modelling of lines, cables and transformers, the knowledge is available, although typical parameters of the capacitance for distribution transformers may not be generally available.

8.2. Device modelling

Devices connected to the grid are the most challenging part of the modelling effort in this frequency range. At the point of connection the characteristic of a device (if the behaviour mimics that of a current or voltage source) significantly depends on the ratio between source impedance of the device and network impedance at the considered frequency. It is even possible that one and the same device behaves at one point more like a voltage source and at another point more like a current source (or source with inner impedance). Detailed models of devices like power electronic converters can be found in the literature, with Refs. [100,101,102] being some recent books on the studied subject. The detail needed to model the behaviour of a device within this frequency range with sufficient accuracy is however still not understood. To model and simulate an installation correctly might mean one has to include all components in the devices connected within the installation and even if that information is available it will become computationally intensive already for one device. In Ref. [104] a comparison is made between a detailed Simulink model and the measured emission. It is shown that the emission can be very well reproduced for lower frequencies, but not for frequencies above 2 kHz. Devices can be modelled as an emitting source but the model also has to consider the fact that every device can in some way impact the propagation of the emission from other sources (see Section 4.3). According to Ref. [58], as cited in Ref. [67], a constant DC voltage behind a voltage-source converter is sufficient as a model of the emitting source. It is shown in Ref. [41] that a simple model of a vacuum cleaner (8 A of current source in parallel with 145 µH inductance and 0.495 µF for the filter capacitor) can reproduce the measurements in the range 2–9 kHz. A battery charger for an electric car is modelled in Ref. [29]; a detailed time-domain model is used for the charger including the EMI filter, the PWM pattern and its control; the grid is modelled by its series impedance. This model was able

to reproduce resonances for a certain cable length, including the 3.4 kHz resonant frequency of the EMI filter.

More detailed modelling of the power–electronics interface of lighting sources is presented among others in Ref. [105]. A complete model of a device also has to cover the complexity of the interaction between devices. There is a linear part concerning the behaviour of a device as an impedance (seen by other devices). There is also a non-linear part: the voltage distortion caused by emission from other grid connected devices, changes the emission of the device. This can be partly addressed by "harmonic fingerprinting" as introduced for (low-frequency) harmonics in Ref. [106] but that would still not cover the interactions between signals at different frequencies.

Models do exist for a range of other components, like inverters for PV panels or wind turbines, adjustable-speed drives, computing equipment and televisions. However, those models have been developed for a lower frequency range and the validity of the models for the frequency range 2–150 kHz needs to be checked.

9. Research challenges

The following research challenges and opportunities for future research have been identified.

9.1. Emission

- Relations between converter configuration and control, EMC filter, and primary emission
- Origin of recurrent oscillations and their aggregation with multiple emitting devices
- Impact of voltage magnitude, distortion, source impedance, and the presence of other end-user equipment nearby, on the primary and secondary emission
- Time-varying nature of the emission

9.2. Propagation

- Transfer between devices and from individual devices to the grid, especially the impact of resonances and damping
- Aggregation of emission from different devices, including the aggregation between primary and secondary emission
- Propagation from sources of emission at medium or high voltage to lower voltage levels
- Distinction between primary and secondary emission by means of modelling and measurements

9.3. Interference

- The impact of end-user equipment (emission and impedance) on the performance of power-line communication
- The impact of short-duration high magnitude emission on end-user equipment
- Emission at new frequencies from non-linear equipment due to background distortion
- Impact of voltage distortion on performance and life length of end-user equipment and equipment in the grid
- Methods for increasing immunity of end-user equipment

9.4. Measurements

- Methods for accurate and reproducible measurement of voltages and currents at higher frequencies
- Methods for extracting time-domain, frequency-domain and time-frequency-domain features from measured waveforms

9.5. Standardization

- Standardized methods for quantifying voltage and current distortion in the range 2–150 kHz
- Emission and immunity limits and standardized tests
- A classification of emission based on phenomena, characteristics of the emission and difference in propagation

9.6. Modelling and simulation

- Models for components in the grid and for end-user equipment connected to the grid
- Models for active converters that can be used to predict their emission, their impact on the emission, and the spread of disturbances from neighbouring devices
- Modelling guidelines for different frequency ranges and different phenomena

10. Conclusions

This paper reviews the literature on voltage and current distortion in the frequency range between 2 and 150 kHz. A considerable amount of material has been published, especially concerning sources of emission and interference in this frequency range. This material should form the basis for a serious and substantial research effort to provide input for further clarification and understanding of its impact on the grid and for control and future standardization. A number of research challenges have been clearly identified and should assist researchers and engineers working with an increasingly complex grid with plenty of sources of higher frequency—supraharmonics.

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